

The Small-World Phenomenon: With Restricted Kleinberg's Model Study

by

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Abstract:

The small-world property, after many years of research, has a well-established theoretical and algorithmic foundation. Among the most notable contributions are Kleinberg's related work and algorithm model. Based on this algorithm model, we added a constraint: long-range contacts can only be generated horizontally or vertically. On a two-dimensional network, we obtained **the optimal r value of 1** under this constraint, and we provided a simple proof under some assumptions. Finally, we analyzed some shortcomings in the proof and used network simulations to preliminarily demonstrate how to address these shortcomings.

1 Introduction

Kleinberg's research provided critical insights into the algorithmic underpinnings of the small-world phenomenon, particularly focusing on how decentralized algorithms can efficiently navigate these networks. He introduced a model that explains how individuals, using only local information, can find short paths in a network that exhibits small-world properties[4]. This work not only addressed the theoretical aspects of network navigation but also had practical implications for designing and understanding systems such as peer-to-peer networks, social networks, and communication networks. Kleinberg's contributions have had a lasting impact on the field, influencing subsequent research on network theory and algorithms.

Next, we will first study his work and then investigate the related conclusions obtained under the constraint that:

Long-range contacts can only be generated horizontally or vertically.

1.1 Background

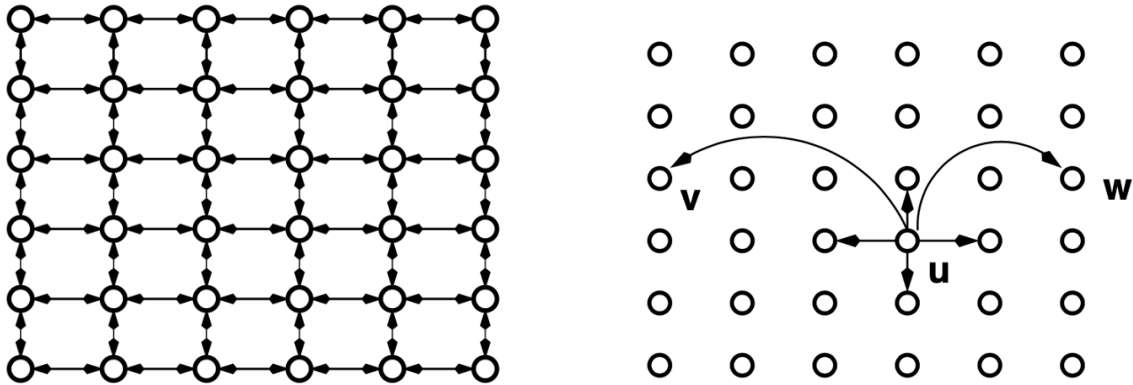
The Small-World Phenomenon

The Small-World Phenomenon refers to the idea that despite the vast number of people in the world, the path connecting any two individuals is relatively short. This phenomenon was first scientifically studied by sociologist Stanley Milgram in the 1960s[1,2,3]. Milgram's experiment, known as the "small-world experiment," involved sending letters through the postal system with the goal of reaching a specific individual, where each recipient was asked to forward the letter to a person they knew who might know the target individual. Milgram found that the median number of steps required to reach the target was around six, known as the "Six Degrees of Separation", highlighting the interconnectedness of social networks.

Kleinberg's Model

We firstly review the network and the model.

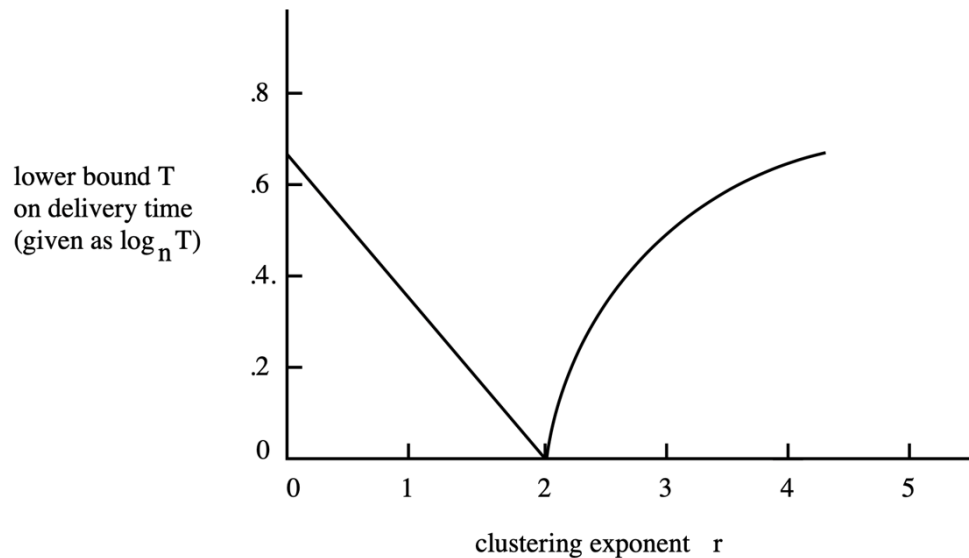
There is the precise definitions for the network model and the notion of a decentralized algorithm[5].



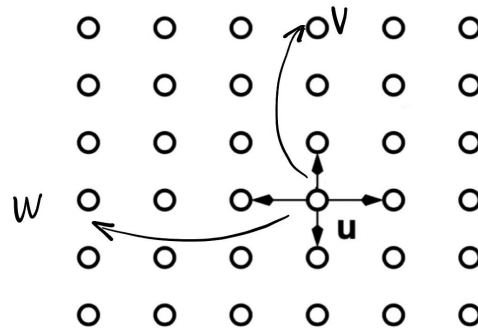
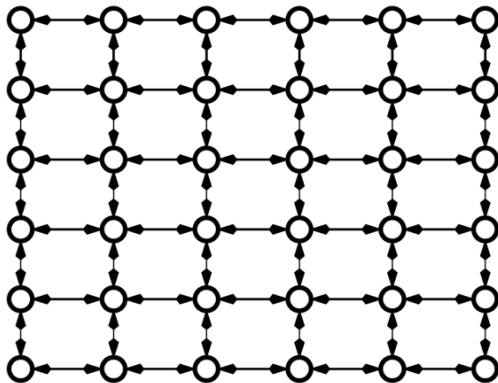
In designing our network model, we seek a simple framework that encapsulates the paradigm of Watts and Strogatz — rich in local connections, with a few long-range connections. Rather than using a ring as the basic structure, however, we begin from a two-dimensional grid and allow for edges to be directed. Thus, we begin with a set of nodes (representing individuals in the social network) that are identified with the set of lattice points in an $n \times n$ square, $\{(i, j) : i \in \{1, 2, \dots, n\}, j \in \{1, 2, \dots, n\}\}$, and we define the lattice distance between two nodes (i, j) and (k, ℓ) to be the number of “lattice steps” separating them: $d((i, j), (k, \ell)) = |k - i| + |\ell - j|$. For a universal constant $p \geq 1$, the node u has a directed edge to every other node within lattice distance p — these are its local contacts. For universal constants $q \geq 0$ and $r \geq 0$, we also construct directed edges from u to q other nodes (the long-range contacts) using independent random trials; the i th directed edge from u has endpoint v with probability proportional to $[d(u, v)]^{-r}$. (To obtain a probability distribution, we divide this quantity by the appropriate nor-

malizing constant $\sum_v [d(u,v)]^{-r}$; we will call this the inverse r th-power distribution.)

The model has the following results:



1.2 This project



We study, as illustrated, the optimal value of r in Kleinberg's Navigation model on a two-dimensional network when long-range contacts can only be generated horizontally or vertically.

We will mimic Kleinberg's modeling process and, for each of his three theorems, provide the corresponding conclusions under the given constraints.

2 Methodology

We first update the theorems under the given constraints, then prove it, and subsequently provide illustrations of the related results.

2.1 Theorems Under Constraint

Theorem 1' *There is a decentralized algorithm A and a constant θ , independent of n , so that when $r = 1$ and $p = q = 1$, the expected delivery time of A is at most $\theta(\log n)^2$.*

Theorem 2' (a) *Let $0 \leq r < 1$. There is a constant θ_r , depending on p, q, r , but independent of n , so that the expected delivery time of any decentralized algorithm is at least $\theta_r n^{(1-r)/2}$.*

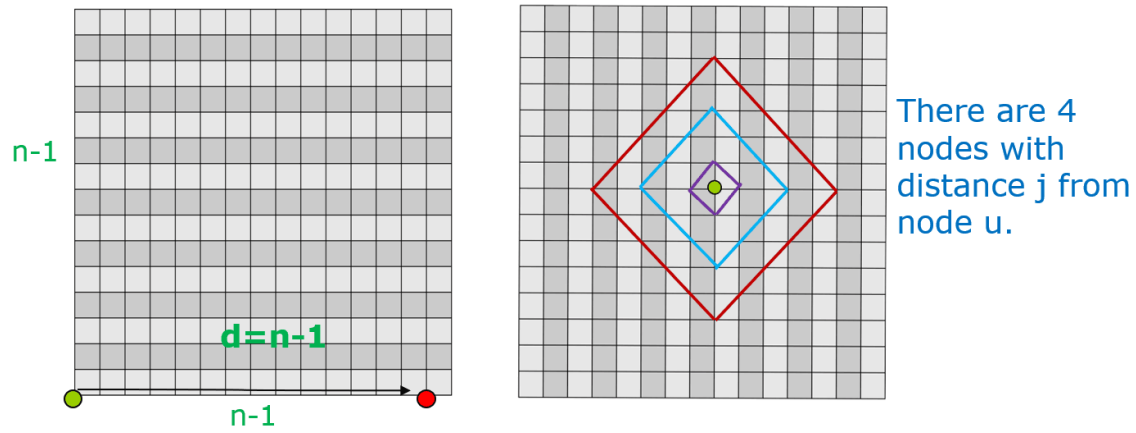
(b) *Let $r > 1$. There is a constant θ_r , depending on p, q, r , but independent of n , so that the expected delivery time of any decentralized algorithm is at least $\theta_r n^{(r-1)/(r)}$.*

2.2 Proof

Following Kleinberg's proof, we use the same symbols and the Principle of Deferred Decisions[8], adding constraints and assumptions where necessary, to provide the following proof:

Proof of Theorem 1': Since $p = q = 1$, we have a network in which each node u is connected to its four nearest neighbors in the lattice (two or three neighbors in the case of nodes on the boundary), and has a single long-range contact v . The probability that u chooses v as its long-range contact is $d(u, v)^{-1} / \sum_{v \neq u} d(u, v)^{-1}$,

and we have[7]:



$$\begin{aligned}
 \sum_{v \neq u} d(u, v)^{-2} &\leq \sum_{j=1}^{n-1} 4 \times j^{-1} \\
 &\leq 4 \sum_{j=2}^{n-1} j^{-1} \\
 &\leq 4 \int_2^n j^{-1} dj \\
 &\leq 4(\ln n - \ln 2) \\
 &\leq 4 \ln n
 \end{aligned}$$

Thus, the probability that v is chosen is at least $[4[\ln(n)]d(u, v)]^{-1}$.

The decentralized algorithm A is defined as follows: in each step, the current message-holder u chooses a contact that is as close to the target t as possible, in the sense of lattice distance, **but only on horizontal or vertical**. For $j > 0$, we say that the execution of A is in *phase* j when the lattice distance from the current node to t is greater than 2^j and at most 2^{j+1} . We say A is in phase 0 when the lattice distance to t is at most 2. Thus, the initial value of j is at most $\log n$.

Suppose we are in phase j , $\log(\log n) \leq j < \log n$, and the current message holder

is u . What is the probability that phase j will end in this step? This requires the message to enter the set B_j of nodes within lattice distance 2^j of t . There are at least

$$1 + \sum_{i=1}^{2^j} i = \frac{1}{2}2^{2j} + \frac{1}{2}2^j + 1 > 2^{2j-1}$$

The above calculations were performed without constraints.

Now, **assuming** these nodes form a square grid (which can be approximated when the number of nodes is large enough), the number of nodes under the constraint condition should be $4B_j^{1/2}$, we denote it as B' .

nodes in B_j , each is within lattice distance $2^{j+1} + 2^j < 2^{j+2}$ of u , and hence each has a probability of at least $(4\ln(n)2^{j+2})^{-1}$ of being the long-range contact of u . If any of these nodes is the long-range contact of u , it will be u 's closest neighbor to t ; thus the message enters B with probability at least

$$\frac{4 \times (2^{2j-1})^{\frac{1}{2}}}{4\ln n \times 2^{j+2}} = \frac{1}{4\sqrt{2}\ln n}$$

Let X_j denote total number of steps spent in phase j , $\log(\log n) \leq j < \log n$. We have

$$\begin{aligned} EX_j &= \sum_{i=1}^{\infty} \Pr[X_j \geq i] \\ &\leq \sum_{i=1}^{\infty} \left(1 - \frac{1}{4\sqrt{2}\ln n}\right)^{i-1} \\ &= 4\sqrt{2}\ln(n). \end{aligned}$$

An analogous set of bounds shows that $EX_j \leq 4\sqrt{2}\ln(n)$ for $j = \log n$ as well. Finally, if $0 \leq j \leq \log(\log n)$, then $EX_j \leq 4\sqrt{2}\ln(n)$ holds for the simple reason that the algorithm can spend at most $\log n$ steps in phase j even if all nodes pass the message to a local contact.

Now, if X denotes the total number of steps spent by the algorithm, we have

$$X = \sum_{j=0}^{\log n} X_j,$$

and so by linearity of expectation we have

$$EX \leq (1 + \log n)(4\sqrt{2}\ln(n)) \leq \theta(\log n)^2$$

for a suitable choice of θ .

Proof of Theorem 2':

Theorem 2' (a)

We consider an arbitrary decentralized algorithm of the type described above, and consider the expected number of steps required for the message to travel from s to t , for nodes s and t generated uniformly at random from the grid.

We need to specify that under the constraint conditions, we do not restrict the total number of points. Points not aligned horizontally or vertically are assumed to be reachable within two steps; this constant factor of 2 does not affect the results.

However, we still constrain the points to be aligned either horizontally or vertically when calculating probabilities.

Note that because we have the freedom to choose the constant θ_r , we may also assume that n is at least as large as some fixed absolute constant n_0 . The probability that a node u chooses v as its i -th out of q long-range contacts is

$d(u, v)^{-r} / \sum_{v \neq u} d(u, v)^{-r}$, and we have

$$\begin{aligned}
 \sum_{v \neq u} d(u, v)^{-r} &\geq \sum_{j=1}^{n/2} 4(j^{-r}) \\
 &= 4 \sum_{j=1}^{n/2} j^{-r} \\
 &\geq 4 \int_1^{n/2} x^{-r} dx \\
 &\geq 4(1-r)^{-1}((n/2)^{1-r} - 1) \\
 &\geq \frac{1}{(1-r)2^{2-r}} \cdot n^{1-r},
 \end{aligned}$$

where the last line follows if we assume $n^{1-r} \geq 2^{2-r}$. Let $\delta = (1-r)/2$.

Let U denote the set of nodes within lattice distance pn^δ of t . Note that

$$|U| \leq 1 + 4 \sum_{j=1}^{pn^\delta} 1 \leq 4pn^\delta,$$

where we assume n is large enough that $pn^\delta \geq 2$. Define $\lambda = (2^{6-r}qp)^{-1}$. Let E' be the event that within λn^δ steps, the message reaches a node other than t with a long-range contact in U . Let E'_i be the event that in step i , the message reaches a node other than t with a long-range contact in U ; thus $E' = \bigcup_{i \leq \lambda n^\delta} E'_i$.

Now, the node reached at step i has q long-range contacts that are generated at random(with constraints) when it is encountered.

So we have

$$\begin{aligned}
\Pr[E'_i] &\leq \frac{q|U|}{[(1-r)2^{2-r}]^{-1} \cdot n^{1-r}} \\
&\leq \frac{(1-r)2^{2-r}q \cdot 4pn^\delta}{n^{1-r}} \\
&= \frac{(1-r)2^{4-r}qpn^\delta}{n^{1-r}}.
\end{aligned}$$

Since the probability of a union of events is bounded by the sum of their probabilities,

$$\begin{aligned}
\Pr[E'] &\leq \sum_{i \leq \lambda n^\delta} \Pr[E'_i] \\
&\leq \frac{(1-r)2^{4-r}\lambda qpn^{2\delta}}{n^{1-r}} \\
&= \frac{(1-r)2^{4-r}\lambda qpn^{2\delta}}{n^{2-r}} \leq \frac{1}{4}.
\end{aligned}$$

We now define two further events. Let $\mathcal{F}$ denote the event that the chosen source s and target t are separated by a lattice distance of at least $n/4$. One can verify that $\Pr[\mathcal{F}] \geq \frac{1}{2}$. Since $\Pr[\overline{\mathcal{F}} \vee E'] \leq \frac{1}{2} + \frac{1}{4}$, $\Pr[\mathcal{F} \wedge \overline{E'}] \geq \frac{1}{4}$.

Finally, let X denote the random variable equal to the number of steps taken for the message to reach t , and let E denote the event that the message reaches t within λn^δ steps.

We claim that if $\mathcal{F}$ occurs and E' does not occur, then E cannot occur. For suppose it does. Since $d(s, t) \geq n/4 > p\lambda n^\delta$, in any $s-t$ path of at most λn^δ steps, the message must be passed at least once from a node to a long-range contact. Moreover, the final time this happens, the long-range contact must lie in U . This contradicts our assumption that E' does not occur.

Thus, $\Pr[E | \mathcal{F} \wedge \overline{E'}] = 0$, hence $E[X | \mathcal{F} \wedge \overline{E'}] \geq \lambda n^\delta$. Since

$$EX \geq E[X|\mathcal{F} \wedge \overline{E'}] \cdot \Pr[\mathcal{F} \wedge \overline{E'}] \geq \frac{1}{4}\lambda n^\delta,$$

part (a) of the theorem follows.

Theorem 2' (b)

We now turn to part (b) of the theorem, when $r > 1$. Again we consider an arbitrary decentralized algorithm; and again, as necessary, we may assume that n is larger than some fixed absolute constant n_0 . We write $\varepsilon = r - 1$. Consider a node u , and let v be a randomly generated long-range contact of u . The normalizing constant for the inverse r -th power distribution is at least 1, and so for any m , we have

$$\begin{aligned} \Pr[d(u, v) > m] &\leq \sum_{j=m+1}^{n-1} 4(j^{-r}) \\ &= 4 \sum_{j=m+1}^{n-1} j^{-r} \\ &\leq \int_m^\infty x^{-r} dx \\ &\leq (r-1)^{-1} m^{1-r} = \varepsilon^{-1} m^{-\varepsilon}. \end{aligned}$$

We set $\beta = \frac{\varepsilon}{1+\varepsilon}$, $\gamma = \frac{1}{1+\varepsilon}$, and $\lambda' = \frac{\min(\varepsilon, 1)}{6q}$. We will assume n has been chosen large enough that $n^\gamma \geq p$. Let E'_i be the event that in step i , the message reaches a node $u \neq t$ that has a long-range contact v satisfying $d(u, v) > n^\gamma$. Let $E' = \bigcup_{i \leq \lambda' n^\beta} E'_i$ be the event that this happens in the first $\lambda' n^\beta$ steps. We have

$$\begin{aligned} \Pr[E'] &\leq \sum_{i \leq \lambda' n^\beta} \Pr[E'_i] \\ &\leq \lambda' n^\beta \cdot q \varepsilon^{-1} n^{-\varepsilon \gamma} \\ &= \lambda' q \varepsilon^{-1} \leq \frac{1}{4}. \end{aligned}$$

As in part (a), we define $\mathcal{F}$ to be the event that s and t are separated by a lattice

distance of at least $n/4$. This is a total lattice distance of at most

$$\lambda' n^{\beta+\gamma} = \lambda' n < \frac{n}{4},$$

Since

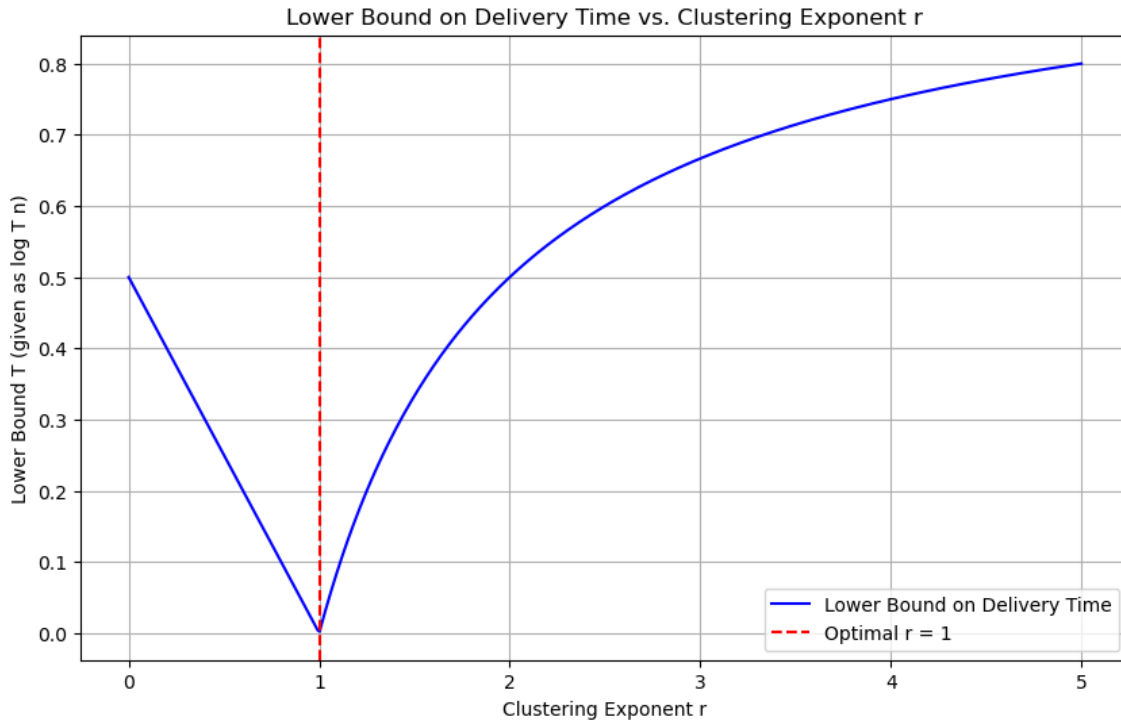
$$EX \geq E[X|\mathcal{F} \wedge \overline{E'}] \cdot \Pr[\mathcal{F} \wedge \overline{E'}] \geq \frac{1}{4} \lambda' n^{\beta},$$

part (b) of the theorem follows.

3 Results and Conclusion

3.1 Results

According the theorem, we have the lower bound implied by Theorem 2'.



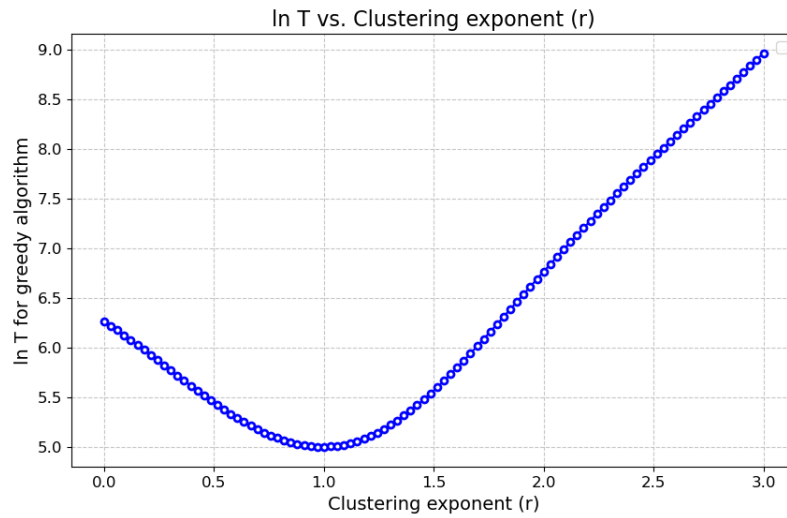
However, we conjecture that when r is sufficiently large, the restriction to horizontal and vertical choices may result in inefficient information propagation over a long period. Because information might be repeatedly and sufficiently propagated within a small area without effectively contributing to reaching the target point.

For this result, we have a natural explanation: because the direction of long-range contacts is restricted, coordinating long-range and short-range links requires limiting the number of short-range links. This increases the probability of choosing the point with the long-range contact. This explains why the value of r here needs to be smaller than the value of r without constraints. Only in this way can long-range and short-range contacts be better coordinated.

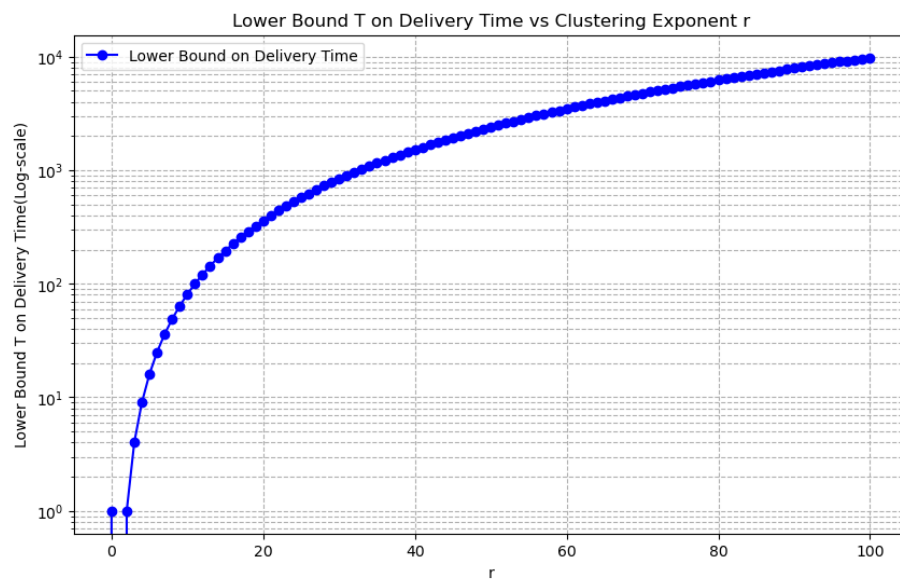
However, this conclusion and proof still have some flaws. If the direction of choosing long-range contacts is restricted, it will inevitably slow down the information transmission speed and may even result in no progress for a long time. In the proof, we ignored the complexity of nodes selection and did not consider the results of multiple repetitions.

Below, we **first** present the simulation results obtained based on the above theorem, and **then** provide a simulation of network navigation under the constraint conditions.

First, we present the results obtained based on the theorem.



Below are the results obtained from network simulations. It is evident that as r increases slightly, the time consumed increases significantly, indicating the inefficiency of information propagation under this constraint condition.



3.2 Conclusion

We imposed constraints on Kleinberg's model and algorithm, restricting long-range contacts to only horizontal or vertical directions. By making some assumptions and mimicking Kleinberg's proof, we provided a simple proof of the theorem under these constraints. However, the real situation is far more complex than described in the theorem. Through simulations and other methods, we preliminarily demonstrated the inefficiency of information propagation under these constraints.

Acknowledgements.

Through the above study and exploration, my understanding of Kleinberg's related work, as well as network navigation and small-world networks, has been enhanced. Although there are related deficiencies, the learning process itself was very enjoyable. Finally, I would like to thank Professor Hu for the semester's teaching, from which I have greatly benefited.

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